Fundamentals of Circuits and Electronics

CSAI 205

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Important policies

1. Lap attendance is Mandatory: Each lab counts; missing a lab will affect your grade.
2. To pass the course, a student must achieve at least **35%** in the final assessment.
3. A minimum of **75%** attendance in both labs and lectures is required for students to pass the course.
4. Any medical excuse must be approved by the **UST clinic,** and the student is responsible for forwarding the clinic's approval to the assigned instructor and teaching assistants within **two** working days. Failure to do so will result in the student being held accountable for any delays, and makeup opportunities may be disregarded.
5. The use of AI tools to complete any assessment is strictly prohibited. Any student found violating this policy will receive a **zero** for the affected assignment and the final course work. This policy is in place to ensure that each student gains a comprehensive understanding of the subject matter through their own efforts and learning. Please adhere to this policy to maintain academic integrity and fair assessment.
6. Please be aware that the attached detailed syllabus content may undergo some modifications throughout the semester. Topics may be added or removed based on the instructor's vision, the collective understanding of the class, and the available time within the semester. Your feedback and engagement will also guide these adjustments to ensure the course meets both learning objectives and time constraints.
7. For any questions or clarifications, students are encouraged to reach out during office hours or via official email. Please allow up to 48 hours for a response, especially for inquiries sent outside of office hours. Please note that communication via WhatsApp or phone calls will **not be accepted**. Official office hours will be announced soon on the classroom platform.

Grading system

	Total number	Grade of each topic	Total weight
Lab practice	12	0.5	6
Quizzes	2	5	10
Projects	6	4	24
Total weight			**40**
Midterm			**20**
Final term			**40**

Main text Books

1. Boylestad, R.L. and Nashelsky, L.,” Electronic Devices and Circuit Theory”, Pearson Education, 2014, ISBN: 9780133109047, Edition: 11.
2. Floyd, T.L.,” Electronic Devices (Conventional Current Version)”, Pearson Education, 2011, ISBN: 9780133004526, Edition: 9.
3. Patrick, D. R., Fardo, S. W., Richardson, R. E., & Chandra, V. V. (2023). *Electronic Devices and Circuit Fundamentals*. River Publishers.

Why this course



Drone = Electronics + Embedded Systems + AI

From sensors and controllers to intelligent decisions



Arduino
Microcontroller

- Simple and flexible
- Great for prototyping
- Used in many DIY drones



Raspberry Pi
Single-board Computer

- Runs Linux
- High-level AI & vision
- Connects to sensors and camera



PIC Microcontrollers

- Low power
- Real-time control
- Reliable and robust



NVIDIA Jetson
AI Edge Computer

- GPU accelerated
- Deep learning inference
- Real-time computer vision





AVR Microcontrollers

- Low power
- High performance
- Widely used in embedded systems



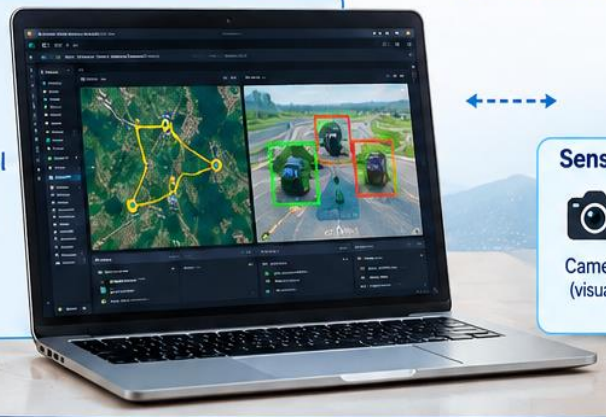
ARM Microcontrollers

- 32-bit architecture
- High speed
- Energy efficient

Communication
(Wi-Fi / Bluetooth / 4G/5G / Radio)

PC / Laptop
AI Software & Mission Control

- Process sensor data
- Run AI models (CV/DL)
- Plan & control the mission
- Visualize results



Sensors

Camera (visual)	Thermal camera	GPS (location)	IMU (position & orientation)	Ultrasonic / LiDAR (obstacle avoidance)

Real-World Applications

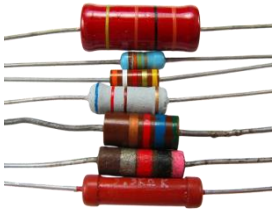
- Agriculture (crop monitoring)
- Surveillance & Security
- Disaster Response
- Smart Cities (traffic, mapping)
- Environmental Monitoring

Electronics + Embedded Systems + AI = Smarter Drones = A Safer, Smarter Future

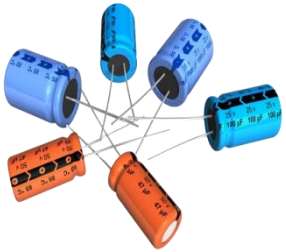
Types of components

Passive

- **Do not need an external power source to operate.**
- **Cannot amplify or generate signals.**



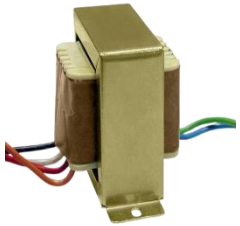
- Resistors



- Capacitors



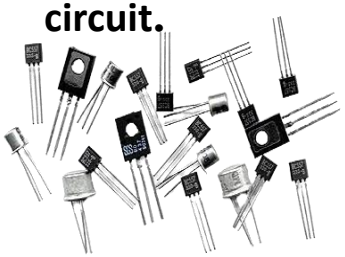
- Inductors



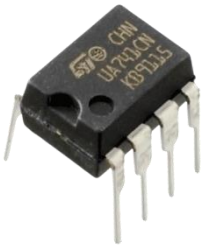
- Transformers

Active

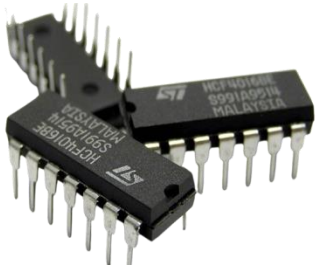
- **Requires external power for operation.**
- **Can generate energy or power within the circuit.**



- Transistors



- Op-Amps



- Integrated Circuits



- Power Supplies

Electricity – The Basics

An understanding of the basics of electricity requires the understanding of three fundamental concepts.

- Voltage
- Current
- Resistance

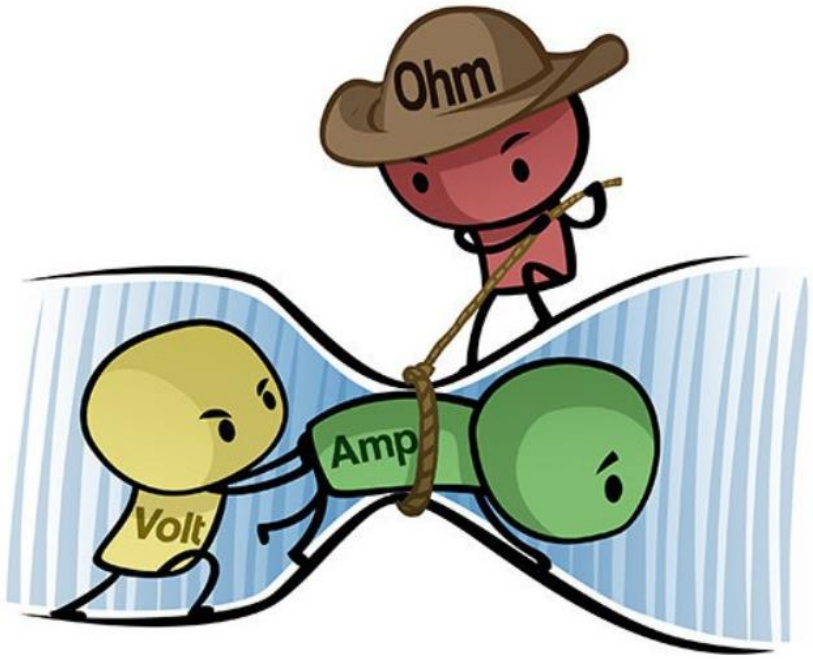
A direct mathematical relationship exists between voltage, resistance, and current in all electronic circuits.

Voltage, Current, & Resistance

Current – Current is the flow of electrical charge through an electronic circuit. The direction of a current is opposite to the direction of electron flow. Current is measured in **AMPERES** (AMPS).

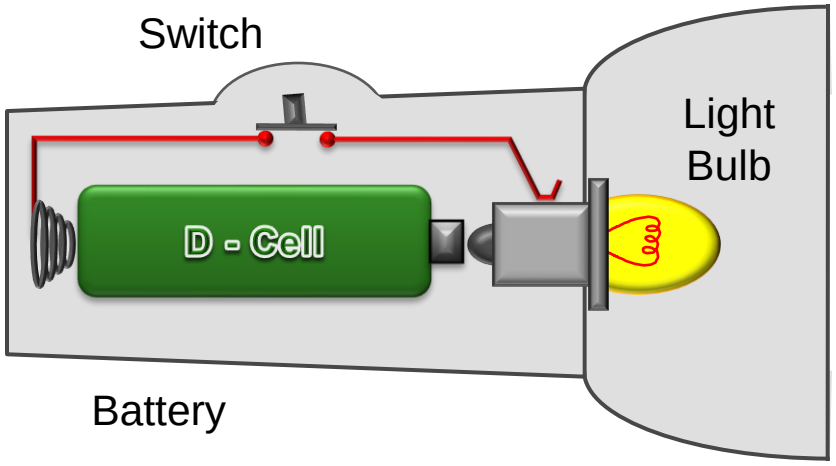
Voltage – Voltage is the electrical force that causes current to flow in a circuit. It is measured in **VOLTS**.

Resistance – Resistance is a measure of opposition to current flow. It is measured in **Ohms**.

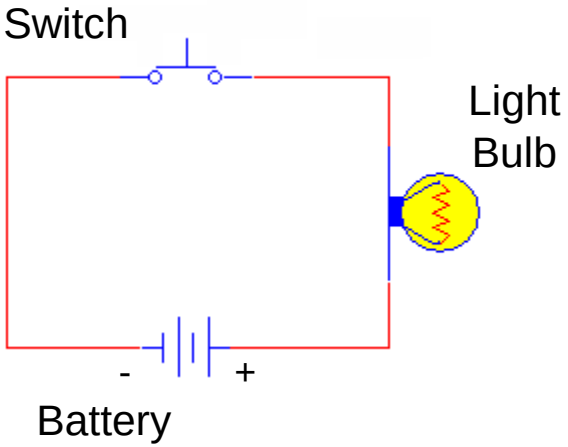


Anatomy of a Flashlight



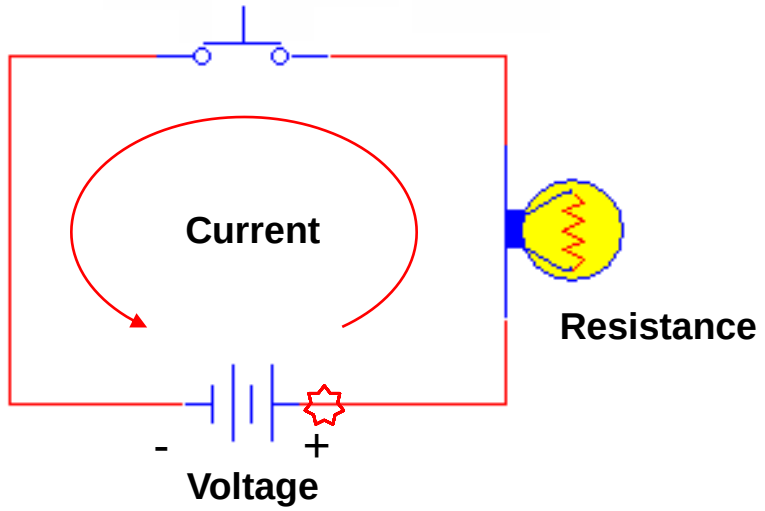


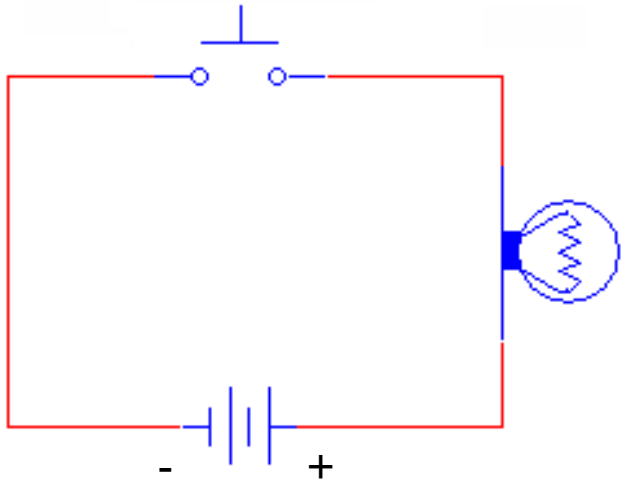
Block Diagram



Schematic Diagram

Flashlight Schematic



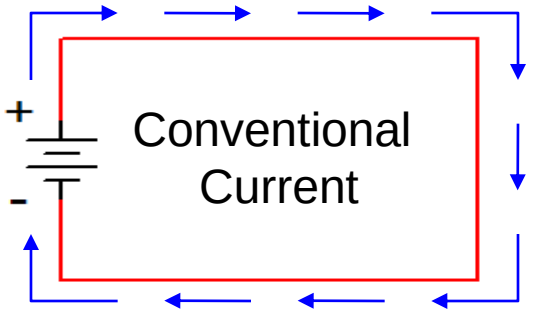


- Closed circuit (switch closed)
- Current flow
- Lamp is on
- Lamp is resistance, uses energy to produce light (and heat)

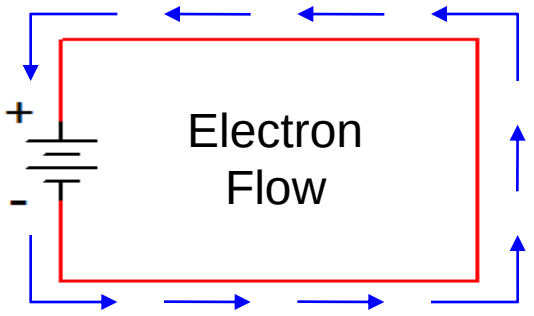
- Open circuit (switch open)
- No current flow
- Lamp is off
- Lamp is resistance, but is not using any energy

Current Flow

- **Conventional Current** assumes that current flows out of the positive side of the battery, through the circuit, and back to the negative side of the battery. This was the convention established when electricity was first discovered, but it is incorrect!

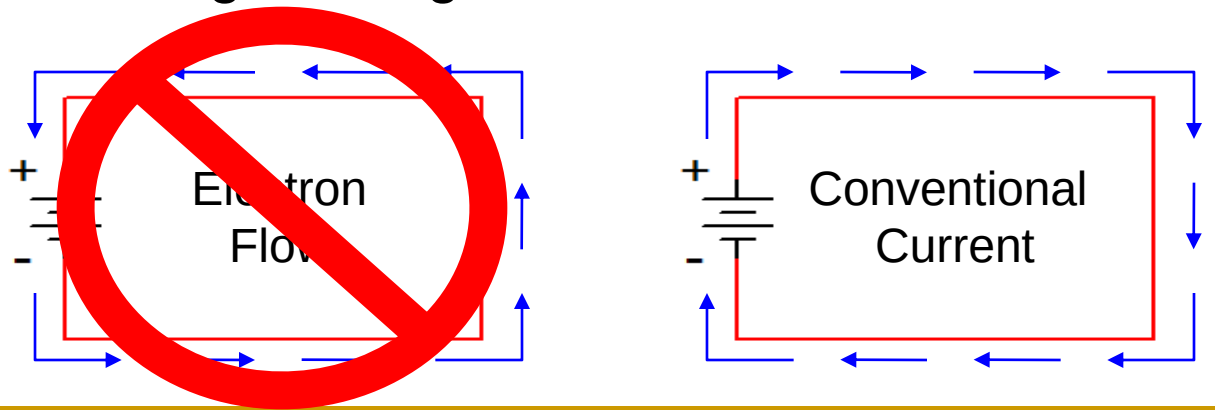


- **Electron Flow** is what actually happens. The electrons flow out of the negative side of the battery, through the circuit, and back to the positive side of the battery.



Engineering vs. Science

- The direction that the current flows does not affect what the current is doing; thus, it doesn't make any difference which convention is used as long as you are consistent.
- Both **Conventional Current** and **Electron Flow** are used. In general, the science disciplines use **Electron Flow,** whereas the engineering disciplines use **Conventional Current.**
- Since this is an engineering course, we will use **Conventional Current**.



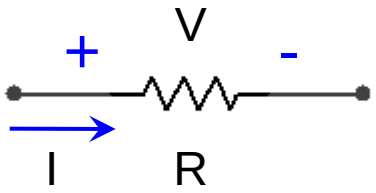
Ohm's Law

- Defines the relationship between voltage, current, and resistance in an electric circuit
- Ohm's Law:

  Current in a resistor varies in direct proportion to the voltage applied to it and is inversely proportional to the resistor's value.

- Stated mathematically:

$$I = \frac{V}{R}$$

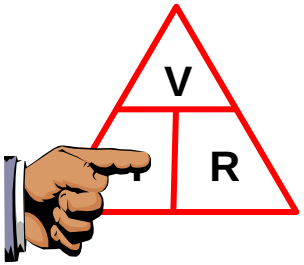


Where: I is the current (amperes)

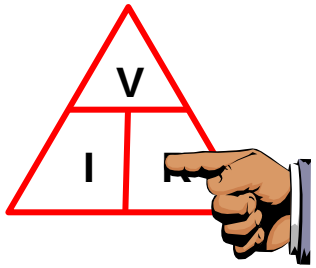
V is the potential difference (volts)

R is the resistance (ohms)

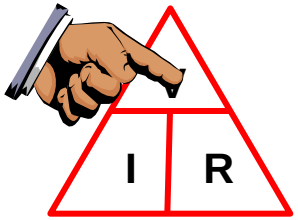
Ohm's Law Triangle



$$I = \frac{V}{R} \quad (\text{amperes}, A)$$



$$R = \frac{V}{I} \quad (\text{ohms}, \Omega)$$



$$V = IR \quad (\text{volts}, V)$$

Example: Ohm's Law

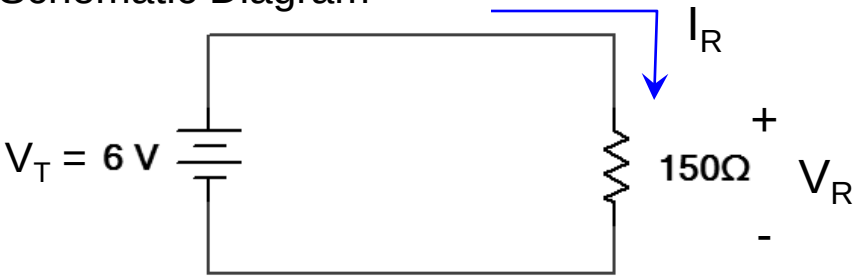


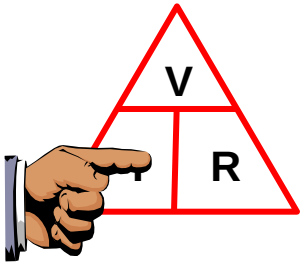
Example:

The flashlight shown uses a 6 volt battery and has a bulb with a resistance of 150 Ω. When the flashlight is on, how much current will be drawn from the battery?

Solution:

Schematic Diagram





$$I_R = \frac{V_R}{R} = \frac{6\,V}{150\,\Omega} = 0.04\,A = 40\,mA$$

Circuit Configuration

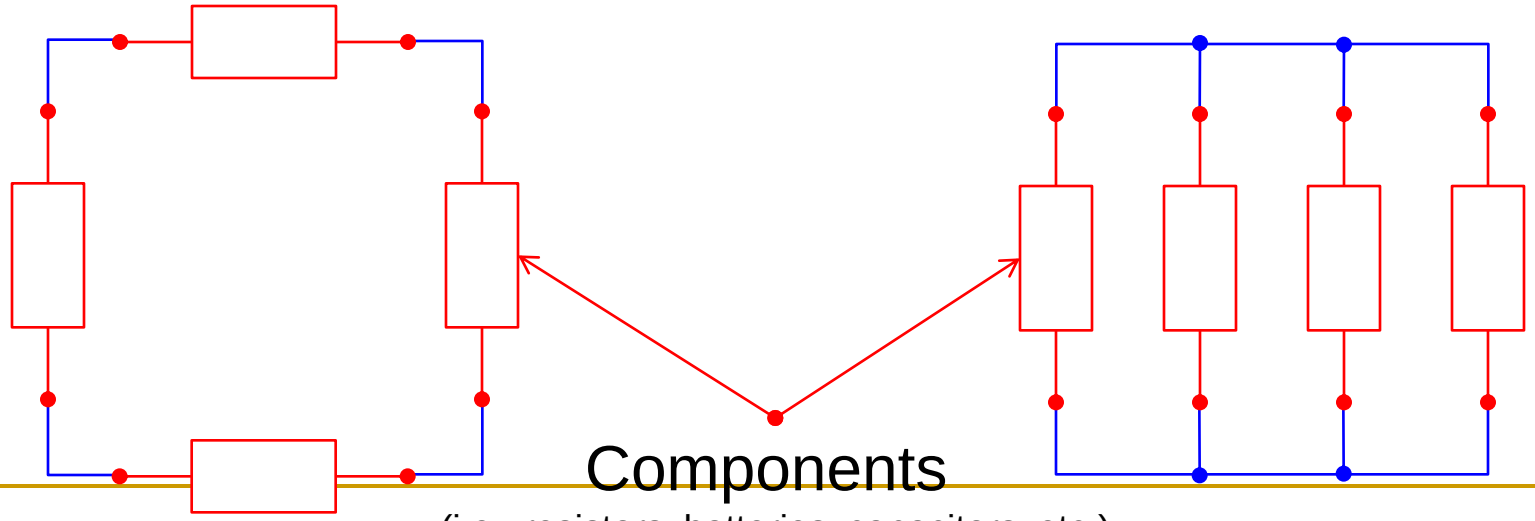
Components in a circuit can be connected in one of two ways.

Series Circuits

- Components are connected end-to-end.
- There is only a single path for current to flow.

Parallel Circuits

- Both ends of the components are connected together.
- There are multiple paths for current to flow.

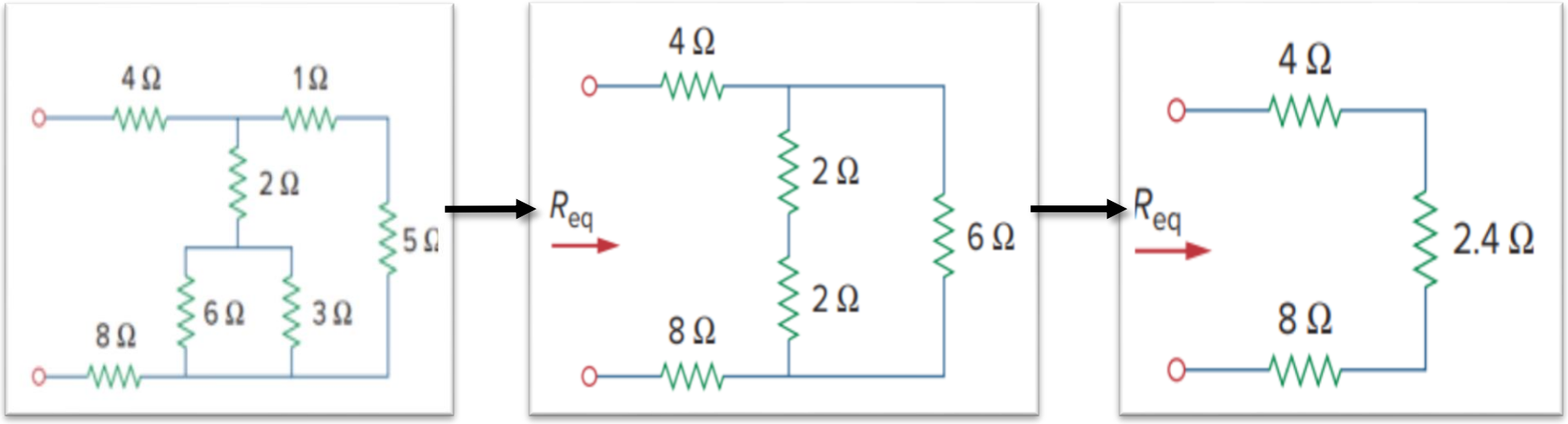


Components

(i.e., resistors, batteries, capacitors, etc.)

Practical Example:

- Find the equivalent resistance for the next figure:



- $1\Omega \; and \; 5\Omega \; are \; Series$:

$$1\Omega + 5\Omega = 6\Omega$$

- $6\Omega \; and \; 3\Omega \; are \; Parallel$:

$$6\Omega \parallel 3\Omega = \frac{6 * 3}{6 + 3} = 2\Omega$$

- $2\Omega \; and \; 2\Omega \; are \; Series$:

$$2\Omega + 2\Omega = 4\Omega$$

- $4\Omega \; and \; 6\Omega \; are \; Parallel$:

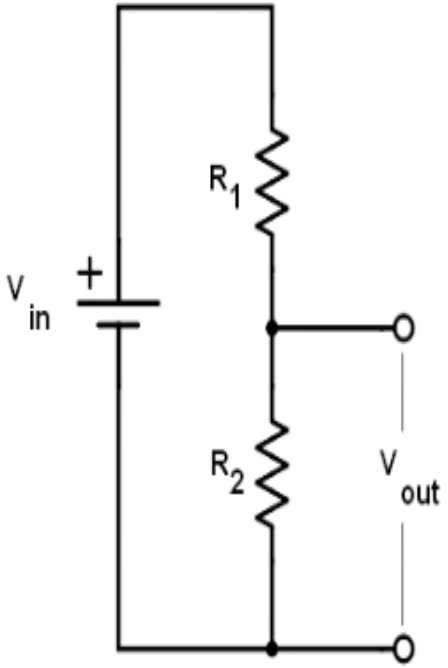
$$4\Omega \parallel 6\Omega = \frac{4 * 6}{4 + 6} = 2.4\Omega$$

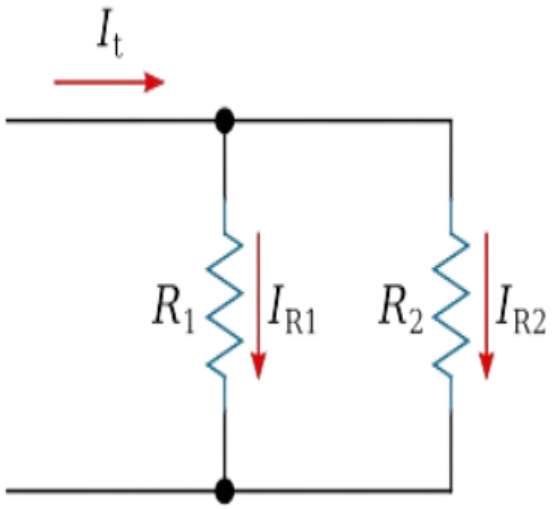
- $4\Omega, 2.4\Omega, \text{and } 8\Omega \; are \; Seri$

$$R_{eq} = 4\Omega + 2.4\Omega + 8\Omega$$

$$= 14.4\Omega$$

Voltage divider & Current divider





Voltage divider

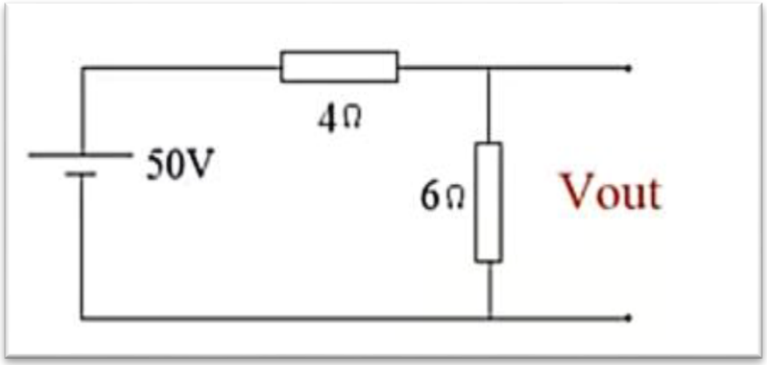
$$V_{out} = V_{in} \frac{R_2}{R_1 + R_2}$$

Current divider

$$I_{R1} = I_t \frac{R_2}{R_1 + R_2} \text{ , } I_{R2} = I_t \frac{R_1}{R_1 + R_2}$$

Practical Examples:

- Determine the value of V_{out}



- Applying Voltage divider law:

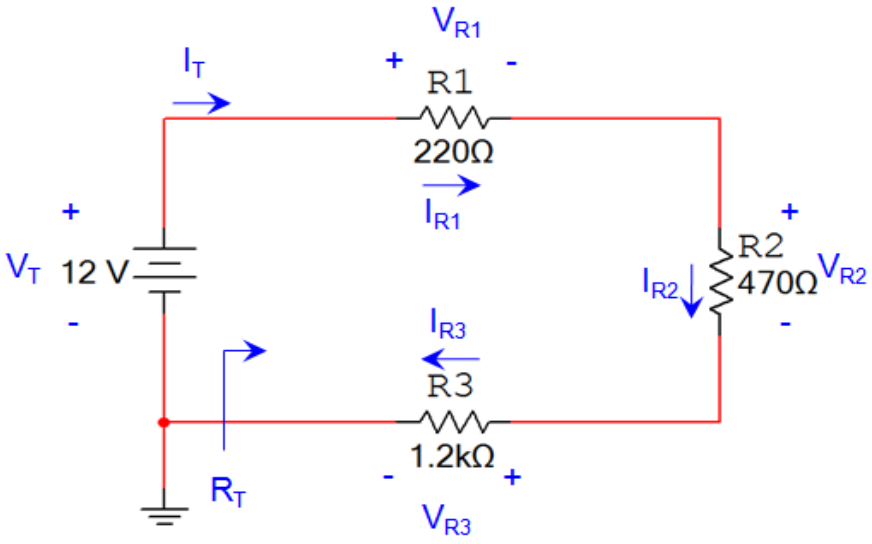
$$V_{out} = 50\frac{6}{6+4} = 30V$$

- Other solution:

$$I = \frac{V}{R_{eq}} = \frac{50}{4+6} = 5A$$

$$V_{out} = IR = 5 * 6 = 30V$$

Example:



Solution:

Total Resistance:

$$R_T = R1 + R2 + R3$$

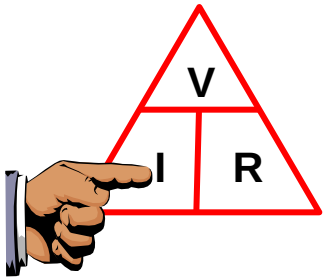
$$R_T = 220\ \Omega + 470\ \Omega + 1.2\ k\Omega$$

$$R_T = 1890\ \Omega = 1.89\ k\Omega$$

Current Through Each Component:

$$I_T = \frac{V_T}{R_T} \quad \text{(Ohm's Law)}$$

$$I_T = \frac{12\ v}{1.89\ k\Omega} = 6.349\ mAmp$$



Since this is a series circuit :

$$I_T = I_{R1} = I_{R2} = I_{R3} = 6.349\ mAmp$$

Voltage Across Each Component:

$$V_{R1} = I_{R1} \times R1 = \quad \text{(Ohm's Law)}$$

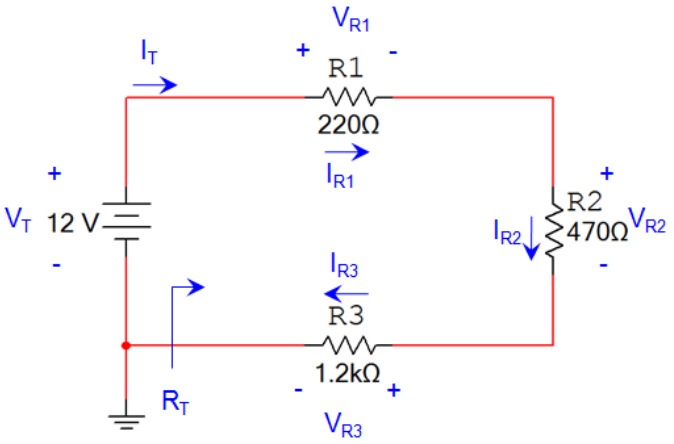
$$V_{R1} = 6.349\,\text{mA} \times 220\,\Omega = 1.397\ \text{volts}$$

$$V_{R2} = I_{R2} \times R2 \ \text{(Ohm's Law)}$$

$$V_{R2} = 6.349\,\text{mA} \times 470\,\Omega = 2.984\ \text{volts}$$

$$V_{R3} = I_{R3} \times R3 \ \text{(Ohm's Law)}$$

$$V_{R3} = 6.349\,\text{mA} \times 1.2\,\text{K}\,\Omega = 7.619\ \text{volts}$$





Verify Kirchhoff's Voltage Law:

$$V_T = V_{R1} + V_{R2} + V_{R3}$$

$$12\,\text{v} = 1.397\,\text{v} + 2.984\,\text{v} + 7.619\,\text{v}$$

$$12\,\text{v} = 12\,\text{v}$$

Example:

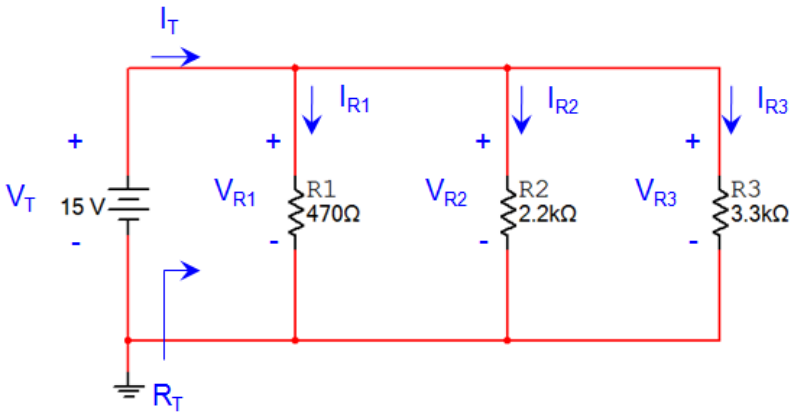
Solution:

Total Resistance:

$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

$$R_T = \frac{1}{\frac{1}{470\,\Omega} + \frac{1}{2.2\,k\Omega} + \frac{1}{3.3\,k\Omega}}$$

$$R_T = 346.59\,\Omega$$



Voltage <u>Across</u> Each Component:

Since this is a parallel circuit:

$V_T = V_{R1} = V_{R2} = V_{R3} = 15$ volts

Current Through Each Component:

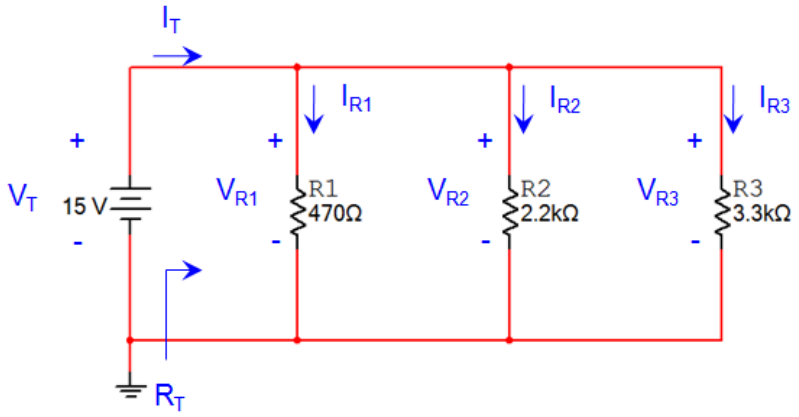
$$I_{R1} = \frac{V_{R1}}{R1} \quad \text{(Ohm's Law)}$$

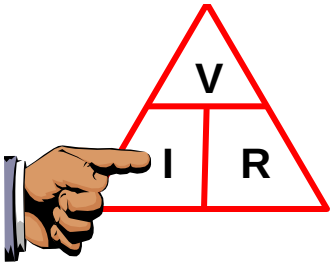
$$I_{R1} = \frac{V_{R1}}{R1} = \frac{15\,\text{v}}{470\,\Omega} = 31.915\,\text{mAmps}$$

$$I_{R2} = \frac{V_{R2}}{R2} = \frac{15\,\text{v}}{2.2\,\text{k}\,\Omega} = 6.818\,\text{mAmps}$$

$$I_{R3} = \frac{V_{R3}}{R3} = \frac{15\,\text{v}}{3.3\,\text{k}\,\Omega} = 4.545\,\text{mAmp}$$

$$I_T = \frac{V_T}{R_T} = \frac{15\,\text{v}}{346.59\,\Omega} = 43.278\,\text{mAmp}$$





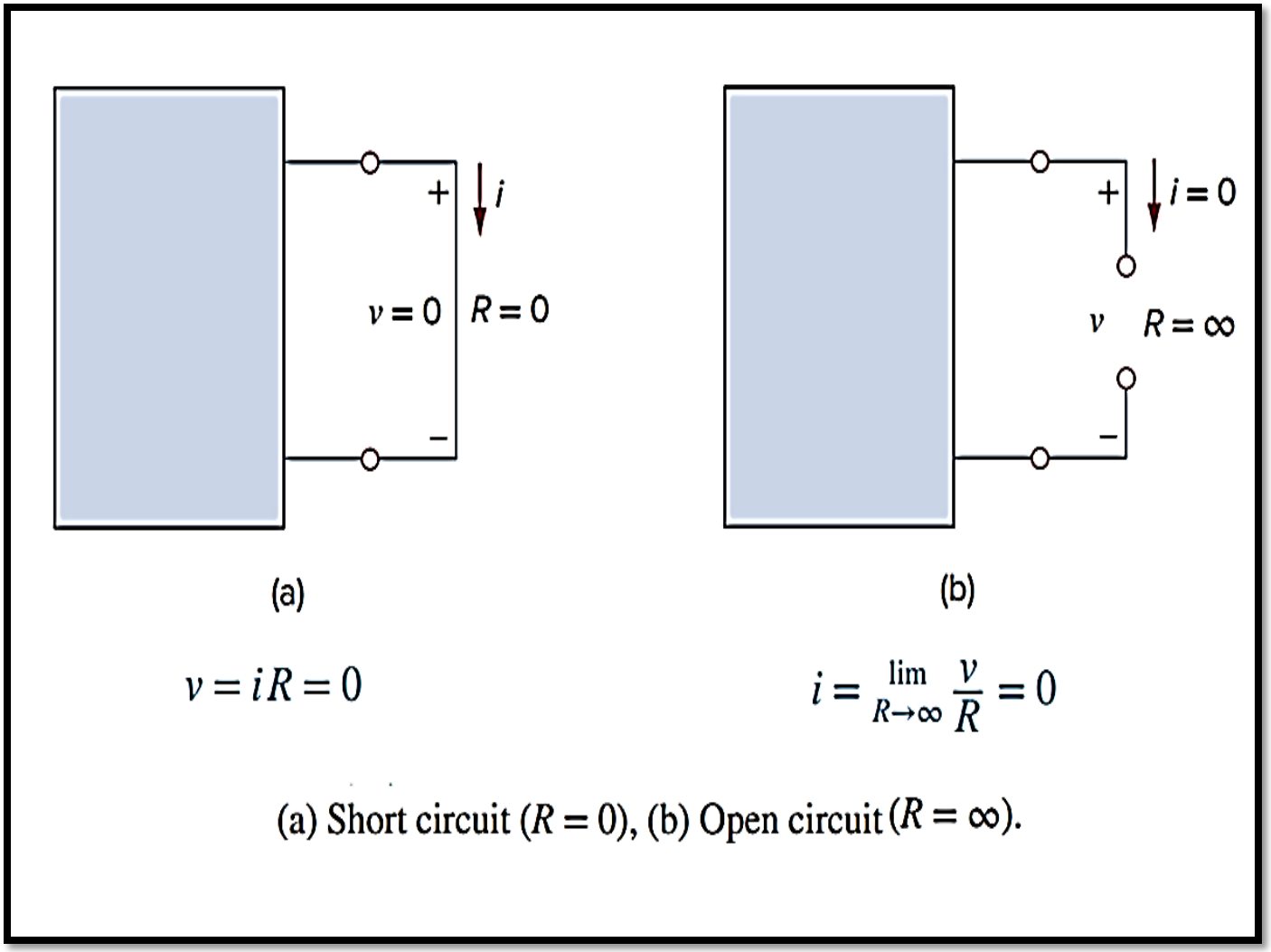
Verify Kirchhoff's Current Law:

$$I_T = I_{R1} + I_{R2} + I_{R3}$$

$$43.278\ \text{mAmps} = 31.915\ \text{mA} + 6.818\ \text{mA} + 4.545\ \text{mA}$$

$$43.278\ \text{mAmps} = 43.278\ \text{mAmps}$$

Special case: Incase of short and open circuits

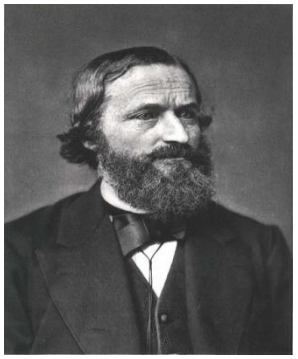


$$v = iR = 0$$

$$i = \lim_{R \to \infty} \frac{v}{R} = 0$$

(a) Short circuit ($R = 0$), (b) Open circuit ($R = \infty$).

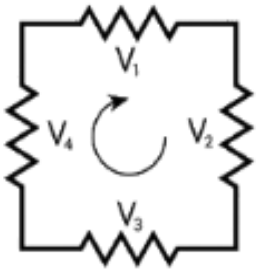
Summary of Kirchhoff's Laws



Gustav Kirchhoff
1824-1887
German Physicist

Kirchhoff's Voltage Law (KVL):

The sum of all of the voltage drops in a series circuit equals the total applied voltage.

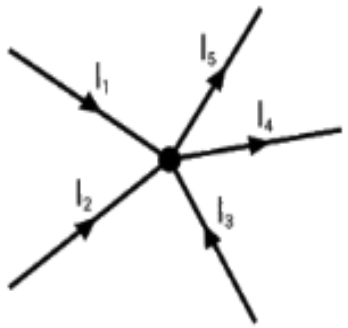


$$\sum V = 0\,, \qquad or \sum V_s = \sum IR$$

$$V_1 + V_2 + V_3 + V_4 = 0$$

Kirchhoff's Current Law (KCL):

The total current in a parallel circuit equals the sum of the individual branch currents.

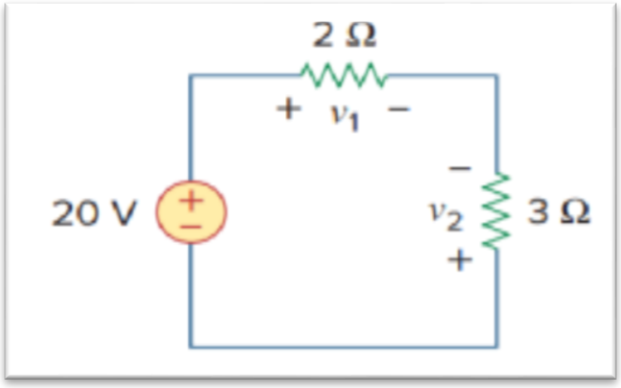


$$\sum I_{in} = \sum I_{out}$$

$$I_1 + I_2 = I_3 + I_4 + I_5$$

Example:

- Find the voltage $V_1, and\ V_2$ for the next circuit :



- Applying Ohm's Law $V = IR$:

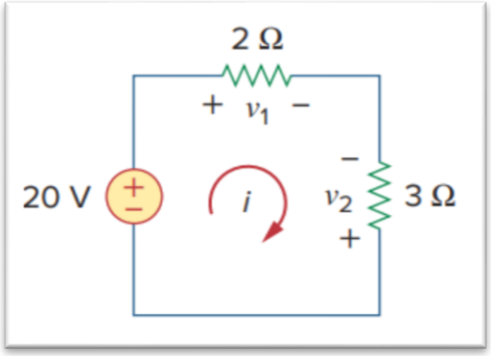
$$v_1 = 2i\ , \qquad v_2 = -3i$$

- Applying KVL on the loop $\sum V = 0$:

$$V_1 - V_2 - 20 = 0$$

$$2i - (-3i) = 20 \ \rightarrow \ 5i = 20v \ \rightarrow i = 4\ A$$

$$v_1 = 2 * 4 = 8V\ , \qquad v_2 = -3 * 4 = -12V$$



Solved Examples

Example (1):

$$+5V - i_1R_1 - i_1R_2 = 0$$

$$5V - i_1(100) - i_1(500) = 0$$

$$5V = i_1(100) + i_1(500)$$

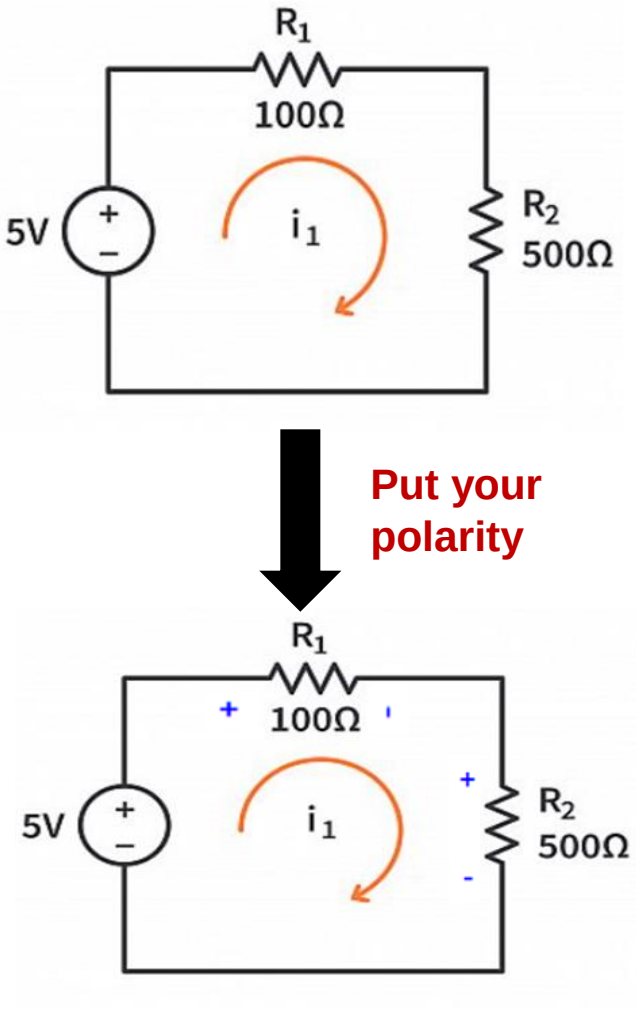
$$5V = 100i_1 + 500i_1$$

$$5V = 600i_1$$

$$i_1 = \frac{5V}{600} = 0.0083A \text{ or } 8.3mA$$

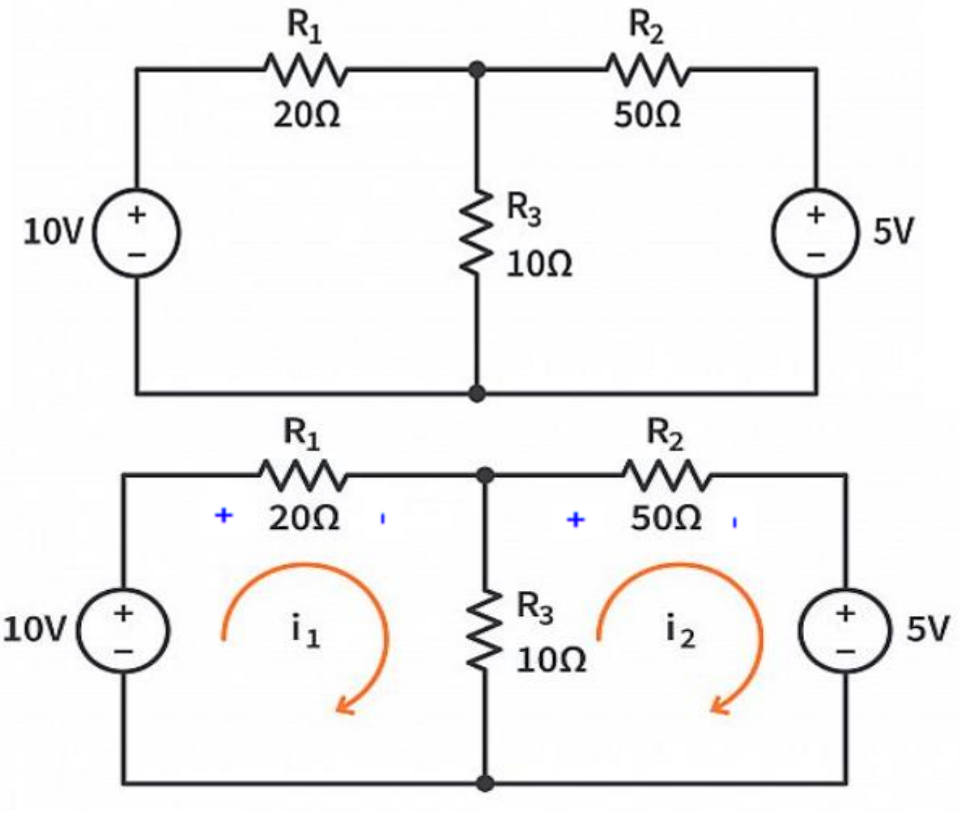
$$V_1 = i_1R_1 = (0.0083)(100) = 0.83V$$

$$V_2 = i_1R_2 = (0.0083)(500) = 4.16V$$



Note that V_1 + V_2 basically equals 5V (rounding errors!) which means that the voltage that drops over the two resistors is the same as the voltage increase from the voltage source.

Example (2):



Loop: 1

$$10\text{V} - \text{i}_1\text{R}_1 - (\text{i}_1 - \text{i}_2)\text{R}_3 = 0$$

$$10\text{V} = \text{i}_1\text{R}_1 + (\text{i}_1 - \text{i}_2)\text{R}_3$$

Loop: 2

$$-(\text{i}_2 - \text{i}_1)\text{R}_3 - \text{i}_2\text{R}_2 - 5\text{V} = 0$$

or

$$-(\text{i}_2 - \text{i}_1)\text{R}_3 - \text{i}_2\text{R}_2 = 5\text{V}$$

Solving two loops together:

$$10\text{V} = \text{i}_1\text{R}_1 + (\text{i}_1 - \text{i}_2)\text{R}_3$$

$$5\text{V} = -(\text{i}_2 - \text{i}_1)\text{R}_3 - \text{i}_2\text{R}_2$$

$$10\text{V} = \text{i}_1(20) + (\text{i}_1 - \text{i}_2)(10)$$

$$5\text{V} = -(\text{i}_2 - \text{i}_1)(10) - \text{i}_2(50)$$

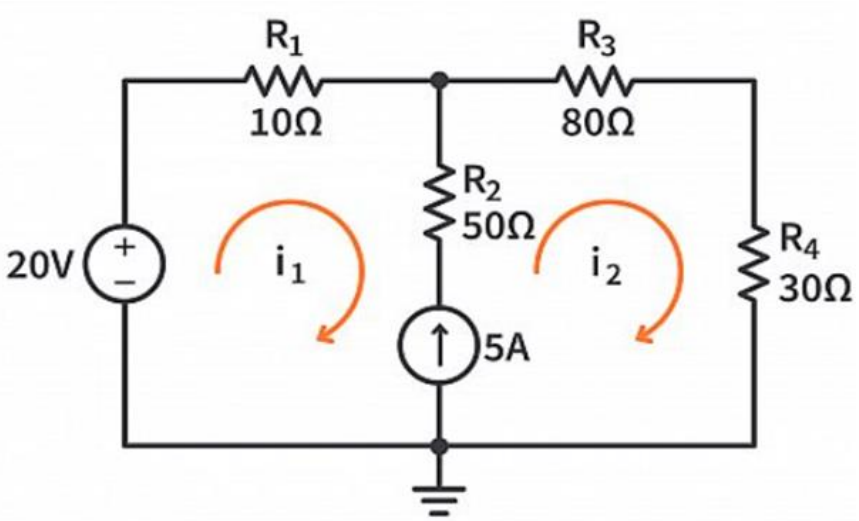
$$10\text{V} = 30\text{i}_1 - 10\text{i}_2$$

$$-60\text{i}_2 + 10\text{i}_1 = 5$$

$$\text{i}_2 = -\frac{5}{170} = -\frac{1}{34} = -0.0294\text{A}$$

$$\text{i}_1 = \frac{1}{3} + \frac{1}{3}(-0.0294) = 0.324$$

Example (3): Supermeshes



From KVL: use the largest loop that doesn't contain current source

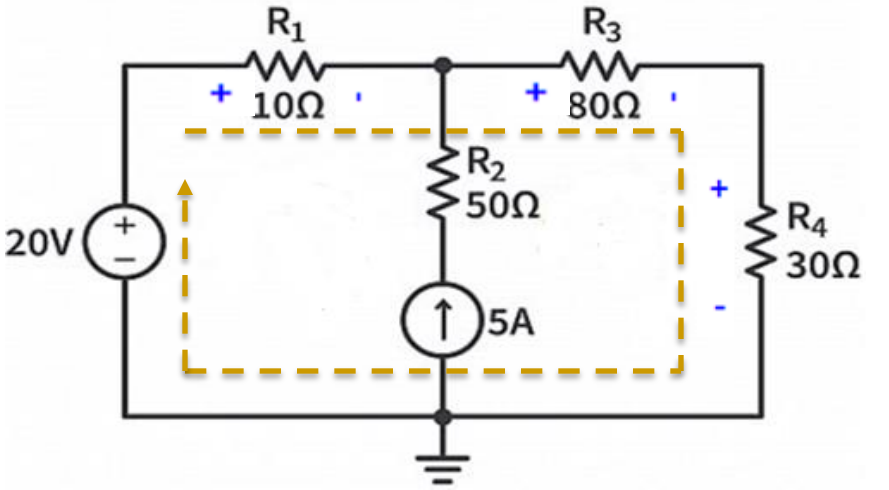
$$+20\text{V} - 10\text{i}_1 - 80\text{i}_2 - 30\text{i}_2 = 0$$

or

$$20 = 10\text{i}_1 + 110\text{i}_2$$

From KCL:

$$5 + \text{i}_1 = \text{i}_2$$



Solving the two equations together:

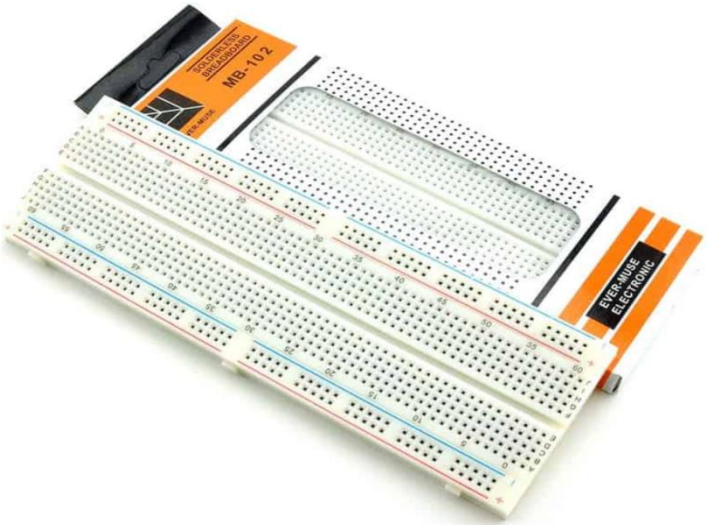
$$10\text{i}_1 + 110\text{i}_2 = 20$$

$$\text{i}_1 - \text{i}_2 = -5$$

$$\text{i}_1 = -\frac{53}{12}\text{A}$$

$$\text{i}_2 = \frac{7}{12}\text{A}$$

Required Tools for practice

Breadboard 830 Tie Point	
Male to Male 20 cm Jumper Wires Set	

Wall Adapter Variable 1.5Vdc to 12Vdc (1A)



Endless Wire DC Female 2.1mm Standard Plug 25 cm.



Basic Model (as shown)

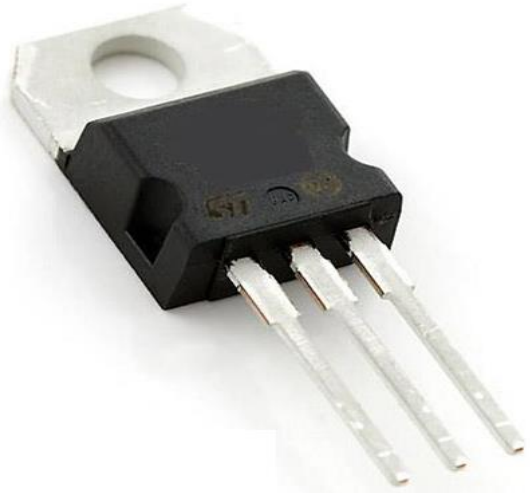
- **DT9205A Digital Multimeter Basic Model MT-1000.**

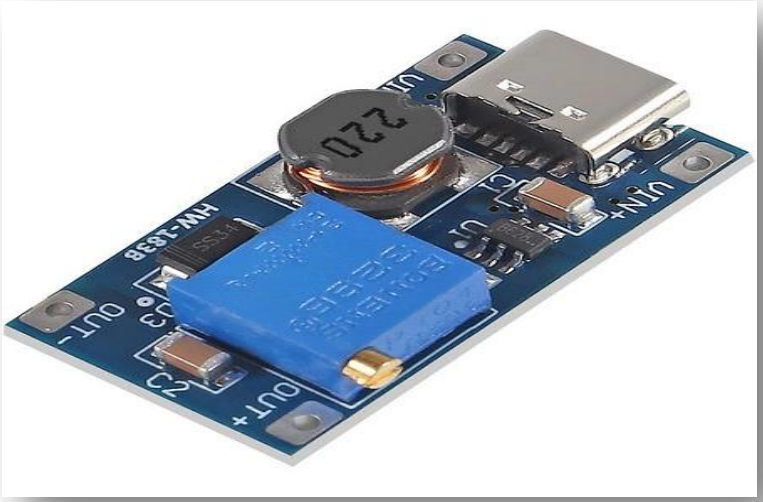
Another Models

- **UT890D+ True RMS Digital Multimeter.**
- **UT39C+ Digital Multimeter.**



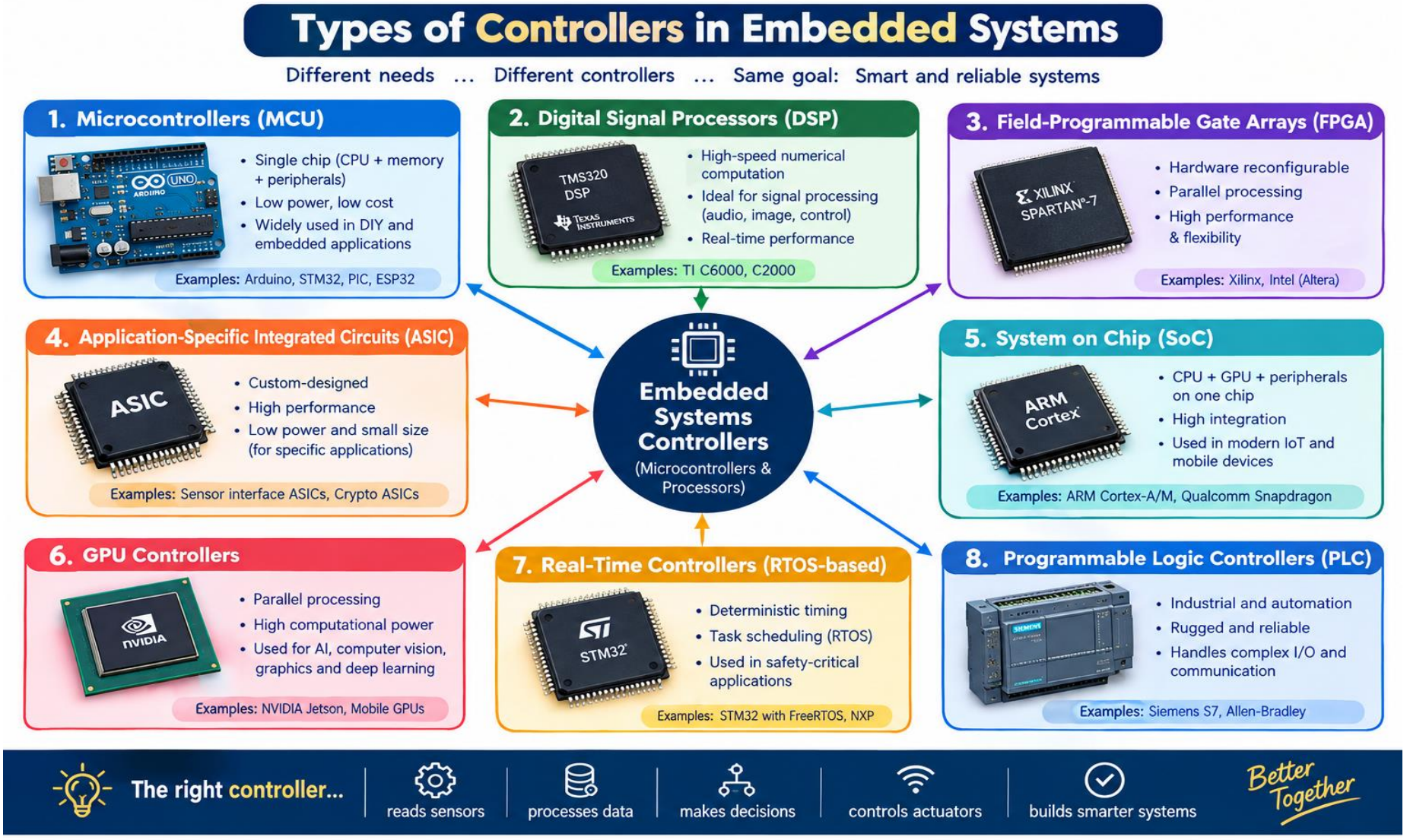
L7805CV & L7809CV & L7812CV (Original ST).



DC Input Regulator 5V Module with DC Jack – Green Version.	
MT3608 DC-DC Step Up Power Boost Module 2A with Type-C USB.	

Advanced level

Controllers



Sensors

